Just Enough to Perform

Knowledge, Provenance, and Epistemic Flattening

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Preamble: The Archive Has Been Flattened into Reach

Provenance is a reconstruction over a suitably deep search space.

We inhabit a peculiar moment in the history of knowledge. What once required months in a library, a run through catalogues, or the patience to chase references through bound journals and microfiche, can now be surfaced in seconds. Claims of provenance that were once practically inaccessible are now recoverable from the deepest searchable archive humanity has yet assembled.

And yet this abundance has not restored depth. It has made flattening easier.

The archive has been flattened into reach. Depth is no longer hidden; it is bypassed.

We say, without hesitation, that from Maxwell's equations we get the speed of light. That statement is not false. It is compressed. It conceals the layered pathway by which a measured ratio, a propagation insight, and a theoretical identification were gradually aligned. Weber measured it, Kirchhoff traced it, Maxwell named it. The result appears timeless; the route by which it became meaningful disappears. This is the new condition of knowledge: not scarcity, but retrieval without reconstruction; not ignorance, but ease. The problem is no longer simply that students do not know enough. It is that they know too much too cheaply, and in forms already polished for repetition.

Knowledge without provenance is indistinguishable from coincidence.

What follows begins from that inversion. Donald Rumsfeld worried that the great danger lay in the unknown unknowns. But in education, the more immediate danger lies elsewhere: in known knowns that have become hollow, publicly available, endlessly repeatable, and detached from the struggle that once made them intelligible.

I. The Known Known-Knowns

The deepest curse of knowledge is forgetting the path, not the destination.

Rumsfeld's taxonomy made ignorance the central danger. The unknown unknown was the apex of risk because one could not even frame the question. But the classroom now presents a different configuration. The danger is not only what cannot be seen. It is what is everywhere seen and nowhere rebuilt.

The modern learner is surrounded by conclusions. Search, summary, prompt, and explanation deliver results at the point of need. Knowledge arrives already compressed into answer form. The result is a culture of what might be called *epistemic hollowness*: the appearance of possession without the labour of construction.

The *curse of knowledge* then deepens. It is not only that the expert in the room assumes others know what they know. It is that the expert no longer remembers how their knowing was laboriously piece-meal assembled. We who are mere teachers, after all. We who occasionally make claims that we may audaciously call our own, arrive at the safe harbour of knowledge and find the drawbridge raised behind us, the path of our arrival erased.

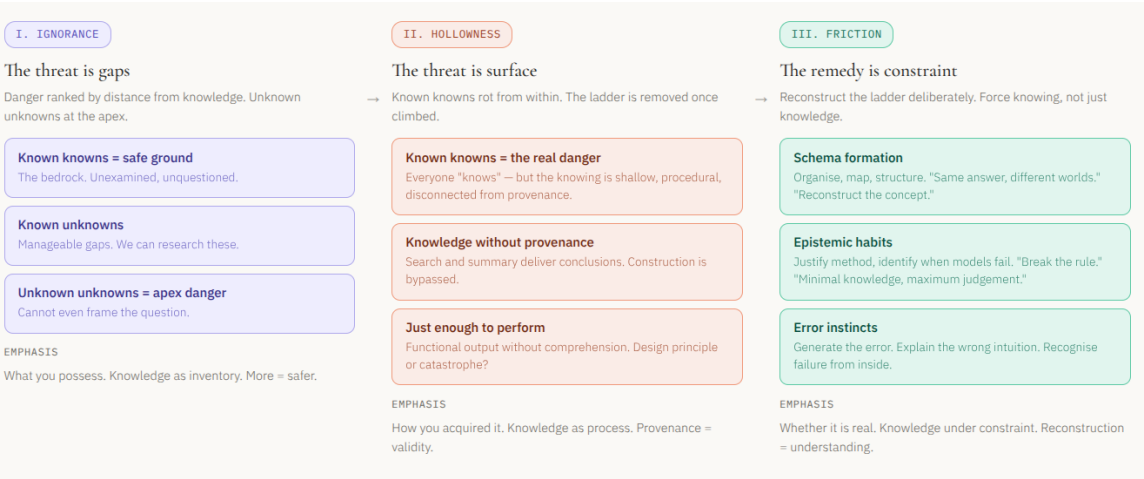

Figure 1: The shift — from ignorance, through hollowness, to deliberate friction. The interactive companion, including the full four-column framework and rubric stress test, is available at leelemacjames.github.io.

II. Knowledge Without Provenance

Retrieval is not reconstruction.

To know something is not merely to hold a statement in memory. It is to inhabit a structure of relation: assumptions, failed starts, rival methods, valid inferences, hidden constraints. When this structure is lost, what remains is portable but thin.

This is why provenance matters. Not because priority disputes are always decisive, nor because intellectual lineage confers automatic authority, but because provenance reveals construction. It restores temporality to what otherwise looks inevitable. It distinguishes measurement from interpretation, result from meaning, inheritance from understanding.

A textbook sentence collapses these distinctions. It turns layered achievement into a single origin point. Depth becomes statement. Process becomes attribution. Contingency becomes inevitability.

Results are path-independent; meaning is not.

Even the most conservative equations harbour a path-dependence.[1]

III. Just Enough to Perform

The danger is not ignorance, but unearned knowledge.

In such a world, education risks being reduced to threshold competence: just enough education to perform. This phrase no longer gestures toward a deficiency in origin; it names a sufficiency at the surface of just enough knowledge to perform, without ever needing to construct.[2] It names a design principle in which the learner need not build the concept if the concept can be summoned procedurally. They need only enough vocabulary, enough structure, enough fluency to output something recognisable as understanding.

[1] In physics, a conservative field is one in which the work done is independent of the path taken. The metaphor here reverses this: the result may be fixed, but the meaning depends on the path by which it is reached.

[2] *Just Enough Education to Perform*, Stereophonics (2001). https://open.spotify.com/album/51I2N3YcrmqOJfzywty3l4?si=OsKPY_8NQHOVVJu9j5l5AA

IV. The Epistemic Cave: Speaking Uncertainty Without feeling

We are witnessing a strange inversion in education.

Students have become fluent in the language of critique passing off knowledge claims, assumptions, counterclaims, methods, and perspectives. They may be able to speak "uncertainty" with some professional precision, but do they rarely have to undergo it.

The Curation of the Self does not breed security

This is the result of a lifetime spent self-editing in text form. When you formulate ideas in a digital buffer rather than on the fly, you lose the vulnerability of the "verbal leap."

In our new "Library Cave," students can state authoritatively that a mathematical method might fail without ever having had to trust one under conditions of risk. They have the vocabulary of the skeptic, but not the scars of the practitioner.

This is why the "Master's Voice" borne of the Scaffolding of Authority in a physical book matters. To build upward in Algebraic Number Theory alone, without a stable lineage is to feel a profound, necessary insecurity.

Knowing that knowledge is uncertain is not the same as being uncertain while knowing.

That distinction is decisive.

V. An Ansatz of Friction

If a student cannot be wrong, they cannot be thinking.

The response cannot be a nostalgic return to encyclopaedic accumulation. Knowledge is abundant, and abundance is not itself the enemy. The problem is frictionless arrival. What is needed, then, is an ansatz of tasks that force reconstruction. Such tasks must cultivate at least three things:

- schemas that organise information,
- habits that define valid reasoning,
- instincts that detect error and overreach.

These are not acquired by exposure alone. They are formed when students must choose between plausible methods, detect tempting mistakes, and discover that a correct answer can conceal invalid reasoning. The task must not merely ask for output; it must expose a method and place that method under strain.

Good tasks withhold closure.

Under such conditions, knowledge becomes more than retrieval. It becomes owned.

VI. A Rubric and a Recursion

What is made assessable becomes optimisable.

Once these epistemic goods are made explicit, however, a recursion emerges. The very act of operationalising them into assessable dimensions turns them into targets. Schema can be performed. Reasoning can be stylised. Error detection can be rehearsed. Epistemic awareness can become a new surface.

This is not a flaw in the rubric so much as its destiny. Every rubric compresses what it measures. It turns lived epistemic tension into observable indicators, and observable indicators can be learned, imitated, and optimised.

Once epistemic awareness is formalised, it becomes performable. Once performable, it becomes detachable from genuine uncertainty.

So the student is no longer reconstructing knowledge, but rehearsing the language of reconstruction.

VII. Epistemic Exposure

Without exposure, critique becomes theatre.

A fourth dimension therefore presses itself upon the earlier triad: epistemic exposure. By this I mean the extent to which a student's reasoning is genuinely vulnerable to being wrong in real time.

This is the part most resistant to assessment and most necessary to understanding. A learner may state assumptions without doubting one, anticipate errors without ever

being surprised, compare methods without ever needing to choose under uncertainty. They may perform doubt fluently while nothing is at stake.

And yet if nothing is at stake, the task has failed epistemically.

Footholds Against Flattening

The greatest obstacle to learning is premature certainty.

If epistemic flattening now abounds in the literature, in pedagogy, and even in the discourse that claims to resist it, then one needs footholds: short, load-bearing sentences that preserve depth against collapse. Not slogans, but reminders of the hidden structure beneath the smooth surface of knowledge. These footholds do not solve the problem. They merely mark where some serious compression has occurred. here they are again.

- Knowledge without provenance is indistinguishable from coincidence.
- Retrieval is not reconstruction.
- The ladder disappears precisely when it has been climbed.
- Familiarity masquerades as understanding.
- What is made assessable becomes optimisable.
- Knowing that knowledge is uncertain is not the same as being uncertain while knowing.
- If a student cannot be wrong, they cannot be thinking.
- Without exposure, critique becomes theatre.

These remind us that the problem is no longer simply ignorance, but surplus certainty; no longer inaccessible depth, but bypassed depth; no longer the absence of knowledge, but its frictionless availability in forms too compressed to inhabit.

To teach now is to restore the conditions under which knowing becomes meaningful. It is to resist the flattening not by withholding access, but by rebuilding depth. It is to make provenance visible again.

To teach is to steepen the incline to lower the drawbridge.

Exercises fit rubrics.
Questions resist them.
Overleaf are some potentially more rewarding Natural Science tasks.

Grab a Foothold

The interactive companion to this paper that includes the full four-column framework, rubric stress test, and epistemic recursion diagram is available at:

`leelemacjames.github.io/epistemic_opacity_full_framework.html`

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames

Nature of Science: Provenance, Models, and Error

We might, in light of the foregoing, develop question tasks that attempt to de-flatten the epistemological incline students now face in the classroom: in science, where models hide assumptions; in mathematics, where patterns hide warrants and mathematics really begins where pattern stops being enough.

And this is all said with that uncomfortable question never far away:

If a question can be fully captured by a marking rubric, is it even worthy of being asked?

Teacher Rationale

This problem set is designed to cultivate three things:

- **Schemas that organise information**: students must connect ideas rather than merely recall facts.
- **Habits that define valid reasoning**: students must justify why a method works and where it does not.
- **Instincts that detect error and overreach**: students must identify tempting mistakes, hidden assumptions, and limits of models.

The tasks are intentially non computational asking students to variously compare, diagnose, reconstruct, and critique.

Instructions to Students

For each question:

- do your own research and show your path to reasoning clearly;
- identify any assumptions you or your sources are making;
- where appropriate, explain why an alternative line of reasoning would fail;
- do not treat a plausible answer as sufficient unless alternative explanations perform the job with fewer assumptions or untestable implications.

Problem 1: Debunking Without Overreach

A claim is made:

> "Humans were hybridised from pigs."

a) List three types of evidence that would be required to support such a claim.

b) Identify which scientific disciplines would be involved.

c) Explain why the absence of evidence is not, by itself, disproof.

d) Construct the strongest possible version of the claim, then dismantle it.

Focus

Not: "this is false"
But: what counts as evidence, and what would it take to know?

Problem 2: Competing Notions of Energy

Compare:

- Buddhist notions of "energy"
- Scientific definitions of energy (work, heat, conservation)

a) Identify what each framework means by "energy."

b) Which claims are testable? Which are not?

c) Is one wrong, or are they operating in different domains?

d) What would count as a meaningful disagreement between them?

Problem 3: Astrology Under Pressure

Astrology assigns significance to constellations such as Sagittarius.

a) Given that Earth precesses and the Sun orbits the galaxy, explain why the original zodiac positions no longer align with current sky positions.

b) If you were an astrologer, how would you revise your model to preserve its predictive power?

c) What would count as falsification of astrology?

d) Identify any aspects of astrology that could be reformulated in a scientifically meaningful way.

Problem 4: What is Matter?

a) Define "inertial mass" in Newtonian terms.

b) Explain how this differs from modern field-based descriptions of matter.

c) If atoms are mostly empty space, in what sense are objects "solid"?

d) Is "billiard-ball atom" a false model or a useful fiction?

Focus

Models may be wrong and still indispensable.

Problem 5: Dead Theories That Worked

Consider:

- Caloric theory of heat
- Phlogiston theory

a) What did each theory get right?

b) Why were they ultimately rejected?

c) Could a theory be wrong and still useful?

d) What current theory might be playing the same role today?

Problem 6: Perpetual Motion and Efficiency

a) Why is 100% efficiency impossible according to thermodynamics?

b) Construct a plausible-sounding perpetual motion argument.

c) Identify precisely where it fails.

d) Why do such ideas remain persuasive?

Problem 7: Models and Reality

George Box said: "All models are wrong, but some are useful."

a) Explain how we can "see molecules" today.

b) Why do we still use Newtonian particle models?

c) Under what conditions is the ideal gas model accurate?

d) Give an example where a simpler model is preferable to a more accurate one.

Problem 8: Galactic Rotation Problem

Stars in galaxies rotate faster than expected.

a) Describe the discrepancy between observation and Newtonian prediction.

b) Option A: modify the law of gravity.

c) Option B: introduce unseen matter.

d) Which would you choose, and why?

e) Give a historical example where science chose one of these strategies.

Problem 9: Inheritance and Stability

Biological traits persist across generations.

a) How can stable traits emerge when molecules are constantly replaced?

b) Why are reproductive organs temperature-sensitive?

c) What does this suggest about the robustness of biological systems?

d) How would you explain this without invoking DNA explicitly?

Problem 10: How vs Why

Consider the question: "Why are leaves green?"

a) Give a mechanistic explanation (how).

b) Give a functional explanation (why).

c) Which is more fundamental?

d) Can a "why" question be answered scientifically?

Problem 11: Perception and Environment

a) Humans are most sensitive to green wavelengths. Why?

b) Why is our radiation strongest at midday?

c) What does our circadian rhythm versus older species suggest about the Earth's rotation?

d) Could life exist without a circadian rythmn?

Problem 12: What is the Ocean?

Eddington asked whether the ocean is made of water or waves.

a) What does each description capture?

b) Are these competing or complementary models?

c) Give another example where two incompatible descriptions are both useful.

Problem 13: The Net Problem

We only observe what our instruments allow.

a) Give an example of a phenomenon that was invisible before new technology.

b) How might current scientific tools limit what we can discover?

c) If we only analyse what we catch, how should we redesign our "nets"?

Final Principle

If your method cannot fail, your knowledge cannot grow.

MYP -friendly Natural Science

Problem 14: The Hidden Assumption in Ohm's Law

A resistor is tested and gives the following results:

Voltage V (V)	Current I (A)	V/I
2	0.5	4
4	1.0	4
6	1.5	4
8	2.0	4

a) What relationship appears to connect voltage and current?

b) What physical quantity seems constant?

c) State the assumption under which this pattern supports Ohm's law.

d) Describe one situation in which this relationship might fail.

e) **Hinge:** Are you observing a law, or fitting a model to limited data?

f) **Extension:** What additional experiment would you design to test whether the relationship truly holds?

Teacher focus

From pattern → law requires assumption.
Students must see where the leap occurs.

Problem 15: Model Failure in Gases

Students use the ideal gas law

$$PV = nRT$$

a) State two assumptions built into the ideal gas model.

b) Explain why the model may become inaccurate at very high pressure.

c) Explain why the model may become inaccurate at very low temperature.

d) Why is a model still useful even when it is not universally true?

e) **Hinge:** At what point does a model stop being "good enough"?

f) **Extension:** Would you rather use a simple wrong model or a complex accurate one? Justify.

Problem 16: Correlation Is Not Causation

A study finds that students who spend more hours revising tend to get higher marks.

a) State what the correlation suggests.

b) Explain why the result does not by itself prove causation.

c) Give one possible confounding variable.

d) Give an example from science where correlation does not imply causation.

e) **Hinge:** What would a convincing causal test look like?

f) **Extension:** Can causation ever be fully proven, or only supported?

Problem 17: Explain the Wrong Intuition

A coin is tossed five times and lands heads each time. A student says:

> "Tails is now more likely on the next toss, because heads has come up too many times already."

a) Explain why this claim feels intuitive.

b) Explain why it is incorrect for a fair coin.

c) State the probability of tails on the next toss.

d) Give a real-life example of similar reasoning.

e) **Hinge:** Why is human intuition poorly adapted to randomness?

f) **Extension:** When is intuition reliable in science, and when is it dangerous?

Problem 18: Measurements and Precision

Two groups measure the boiling point of a liquid.
Group A: 99.8°C, 100.2°C, 100.1°C
Group B: 97.4°C, 102.6°C, 100.0°C

a) Compute the mean for each group.

b) Which group appears more precise?

c) Which group appears more accurate?

d) Explain the difference between precision and accuracy.

e) **Hinge:** Which would you trust more in a new experiment, and why?

f) **Extension:** Can a result be precise, accurate, and still wrong?

Problem 19: Classification and Boundary Cases

Biologists often classify objects into groups, but some cases are awkward.
Consider the categories:

- living and non-living;
- solid and liquid;
- metal and non-metal.

a) Give one example of a boundary case.

b) Explain what this reveals about classification.

c) Why are definitions often imperfect?

d) **Hinge:** Does nature have sharp boundaries, or do we impose them?

e) **Extension:** When do vague definitions become scientifically problematic?

Mixed Epistemic Tasks

Problem 20: The Plausible Graph

A graph shows a quantity rising steeply for three days. A student says:

> "This proves the quantity will continue rising at the same rate for the rest of the month."

a) Why is the conclusion an overreach?

b) What further data would you need?

c) Give one misleading real-world example.

d) **Hinge:** When does a trend become a law?

e) **Extension:** Why do humans over-extrapolate from small datasets?

Problem 21: Rank the Claims

1. Triangle angles sum to 180°
2. Medicine reduced symptoms
3. Correlation implies causation
4. Rule works for three examples

a) Rank the claims.

b) Justify your ranking.

c) **Hinge:** What makes a claim "secure"?

d) **Extension:** Could the ranking change under different assumptions?

Problem 22: Knowledge with Provenance

Choose **one** theorem, law, or rule from mathematics or science that you know well.

Examples:

- Pythagoras' theorem
- conservation of energy, momentum or charge
- Newton's second law or law of gravitation
- Snell's law of refraction or the law of reflection

a) State the rule.

b) Describe its supporting structure.

c) Identify assumptions.

d) Give a common misunderstanding.

e) Explain why provenance matters.

f) **Hinge:** Would you trust a result more if you derived it yourself?

g) **Extension:** Is knowledge ever fully separable from how it was obtained?

Mathematics

Problem 23: Reconstruct the Rule

Consider the sequence

$$2,\ 5,\ 10,\ 17,\ 26,\ 37, \ldots$$

a) Find a formula for the nth term.

b) Give two different ways of seeing the structure of the sequence.

c) A student says, "The differences are odd numbers, so the formula must be exponential." Explain the error.

d) Write a short note describing what type of pattern would lead you to suspect a quadratic rule.

e) **Hinge:** At what point do you move from noticing a pattern to claiming a general rule?

f) **Extension:** Can a finite list of terms ever determine a sequence uniquely? Explain.

Teacher focus

From pattern recognition to rule formation.
Students should feel the gap between "it seems" and "it must be".

Problem 24: Invent the Error

A student solves

$$\sqrt{x^2} = x$$

and concludes that the expression always simplifies to x.

a) Construct the most convincing wrong argument you can for this conclusion.

b) Then explain precisely why it fails.

c) State the correct simplification.

d) Give one numerical example where the wrong rule breaks.

e) **Hinge:** Why do algebraic errors often look more convincing when the notation is familiar?

f) **Extension:** Is an error more dangerous when it gives the wrong answer, or when it sometimes gives the right one?

Teacher focus

Error is not just failure; it reveals the hidden assumptions students are making.

Problem 25: When Does the Rule Break?

A class has learned that

$$(a+b)^2 = a^2 + 2ab + b^2.$$

A student then claims that

$$\sqrt{a+b} = \sqrt{a} + \sqrt{b}.$$

a) Test the claim with $a = 4$ and $b = 5$.

b) Why might the claim seem plausible?

c) Describe a general strategy for checking whether an algebraic identity is true.

d) Give one more false but tempting algebraic statement of your own, and explain why a student might believe it.

e) **Hinge:** What makes one symbolic pattern legitimate and another illegitimate?

f) **Extension:** Is checking a few examples enough to refute a universal algebraic claim? Is it enough to prove one?

Teacher focus

Students should distinguish resemblance from rule.
Symbolic familiarity is not mathematical validity.

Problem 26: Translate Across Contexts

Each of the following situations may be modelled by a similar mathematical form:

- compound interest in a bank account;
- population growth in a favourable environment;
- radioactive decay.

a) Identify the common mathematical structure.

b) Explain what differs in interpretation across the three contexts.

c) Which of the three contexts involves a decreasing quantity rather than an increasing one?

d) Why is it valuable to recognise the common structure beneath different stories?

e) **Hinge:** What is gained, and what is lost, when very different situations are described by the same mathematics?

f) **Extension:** When does a shared mathematical form reveal a deep connection, and when is it only a superficial similarity?

Teacher focus

Abstraction is powerful, but students should ask what the abstraction leaves out.

Problem 27: Averages and Overreach

A student scores 40, 60, and 80 on three tests. Another student scores 55, 60, and 65.

a) Find the mean score for each student.

b) Find the range for each student.

c) Which student is "better"? Explain why this question cannot be answered from the mean alone.

d) Describe a real situation in science or public policy where relying on a single average could be misleading.

e) **Hinge:** Why do averages feel more objective than they really are?

f) **Extension:** Can a summary statistic clarify and distort at the same time? Give an example.

Teacher focus

A summary is not the thing itself.
Students should see how compression can hide structure.

Optional Reflection

1. When did you feel most uncertain in your reasoning?
2. Which question forced you to rethink your answer?
3. What would count as being wrong in your strongest answer?